

On Consistency of Optimal Matching-based clustering algorithm for Markov Chains

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1 Convergence of the OM dissimilarity

1.1 Definitions

We collect here the main objects and notations used in the sequel.

Definition 1 (Alphabet and strings). *Let Σ be a finite alphabet. For $n \in \mathbb{N}$, we write Σ^n for the set of strings $x_{1:n} = (x_1, \dots, x_n)$ with letters in Σ , and $\Sigma^{\mathbb{N}}$ for the set of one-sided sequences $x = (x_t)_{t \geq 1}$.*

Definition 2 (Optimal Matching (edit) dissimilarity). *Fix a symmetric substitution cost scheme $c_{\text{sub}} : \Sigma \times \Sigma \rightarrow [0, \infty)$ with $\bar{c}(a, a) = 0$, insertion and deletion costs $c_{\text{ins}}, c_{\text{del}} : \Sigma \rightarrow (0, \infty)$. Let $\Sigma_{\perp} := \Sigma \cup \{\perp\}$, where $\perp$ denotes a gap symbol.*

For two strings $x_{1:n} \in \Sigma^n$ and $y_{1:m} \in \Sigma^m$, an edit path from $x_{1:n}$ to $y_{1:m}$ is a finite sequence of elementary operations applied successively:

- substitute a letter $a \in \Sigma$ by $b \in \Sigma$ (cost $c_{\text{sub}}(a, b)$);
- delete a letter $a \in \Sigma$ (cost $c_{\text{del}}(a)$);
- insert a letter $b \in \Sigma$ (cost $c_{\text{ins}}(b)$).

The cost of an edit path is the sum of its operation costs, and the Optimal Matching (OM) distance is defined by

$$d_{\text{OM}}(x_{1:n}, y_{1:m}) := \min\{\text{cost}(\pi) : \pi \text{ edit path transforming } x_{1:n} \text{ into } y_{1:m}\}.$$

Equivalently, one may optimize over alignments. An alignment of $(x_{1:n}, y_{1:m})$ is a pair of strings $(\tilde{x}_{1:L}, \tilde{y}_{1:L}) \in \Sigma_{\perp}^L \times \Sigma_{\perp}^L$ such that

- (i) *for all $\ell \in \{1, \dots, L\}$, $(\tilde{x}_{\ell}, \tilde{y}_{\ell}) \neq (\perp, \perp)$;*
- (ii) *removing the $\perp$ symbols from $\tilde{x}_{1:L}$ yields $x_{1:n}$, and removing the $\perp$ symbols from $\tilde{y}_{1:L}$ yields $y_{1:m}$.*

The cost of an alignment is

$$\text{cost}(\tilde{x}, \tilde{y}) := \sum_{\ell=1}^L \begin{cases} c_{\text{sub}}(\tilde{x}_\ell, \tilde{y}_\ell), & \text{if } \tilde{x}_\ell \neq \perp, \tilde{y}_\ell \neq \perp, \\ c_{\text{del}}(\tilde{x}_\ell), & \text{if } \tilde{x}_\ell \neq \perp, \tilde{y}_\ell = \perp, \\ c_{\text{ins}}(\tilde{y}_\ell), & \text{if } \tilde{x}_\ell = \perp, \tilde{y}_\ell \neq \perp. \end{cases}$$

The minimal alignment cost equals $d_{\text{OM}}(x_{1:n}, y_{1:m})$:

$$d_{\text{OM}}(x_{1:n}, y_{1:m}) := \min \{ \text{cost}(\tilde{x}, \tilde{y}) : (\tilde{x}, \tilde{y}) \text{ alignment of } (x_{1:n}, y_{1:m}) \}.$$

The equivalence between the OM dissimilarity and the optimal alignment search is a well known result, see [OO13].

Assumption 1 (Metric cost scheme). *Throughout the rest of the paper, we assume that the cost functions $(c_{\text{ins}}, c_{\text{del}}, c_{\text{sub}})$ satisfy:*

- (i) $c_{\text{ins}}(a) = c_{\text{del}}(a) =: \delta(a)$ for all $a \in \Sigma$,
 - (ii) $c_{\text{sub}}(a, b) = c_{\text{sub}}(b, a)$ for all $a, b \in \Sigma$,
 - (iii) $c_{\text{sub}}(a, c) \leq c_{\text{sub}}(a, b) + c_{\text{sub}}(b, c)$ for all $a, b, c \in \Sigma$.
- Under these conditions, d_{OM} defines a metric on $\Sigma^{\mathbb{N}}$.

1.2 OM convergence

Let $X = (X_t)_{t \geq 1}, Y = (Y_t)_{t \geq 1}$ be discrete-time stationary ergodic processes on a finite alphabet Σ , and $\bar{c}$ a symmetric substitution cost scheme on Σ . We denote d_{OM} the Optimal Matching distance on $\mathcal{X}$ generated by c_Σ and insertion/deletion costs for each $s \in \Sigma$, and $D_n := d_{\text{OM}}(X_{1:n}, Y_{1:n})$.

Proposition 1. $\left(\frac{D_n}{n}\right)$ converges almost surely, and its limit $\gamma(X, Y)$ only depends on $(\mathcal{L}(X), \mathcal{L}(Y))$.

Proof. Let $n, m \in \mathbb{N}$ and let G be an optimal alignment between $x_{1:n+m}$ and $y_{1:n+m}$. Let G_1 and G_2 be optimal alignments between $x_{1:n}$ and $y_{1:n}$, and between $x_{n+1:n+m}$ and $y_{n+1:n+m}$, respectively. Consider the concatenated alignment

$$\underbrace{(1, \dots, n)}_{G_1} \quad \underbrace{(n+1, \dots, n+m)}_{G_2}.$$

Its cost is $c(G_1) + c(G_2)$. Since G is optimal on $x_{1:n+m}$ and $y_{1:n+m}$, we have $c(G) \leq c(G_1) + c(G_2)$. Hence,

$$d_{\text{OM}}(x_{1:n+m}, y_{1:n+m}) \leq d_{\text{OM}}(x_{1:n}, y_{1:n}) + d_{\text{OM}}(x_{n+1:n+m}, y_{n+1:n+m}).$$

Equivalently, $D_{n+m} \leq D_n + D_m \circ T^n$, where T denotes the shift operator on $\mathcal{X}$.

It then follows from Kingman's subadditive ergodic theorem [Ste89] that (NB : quelle mesure mettre, qui joue le rôle de g_n , lemme de Fekete)

$$\frac{D_n}{n} \xrightarrow[n \rightarrow \infty]{\text{a.s.}} \gamma,$$

where γ depends only on the joint law of (X, Y) . □

Remark 1. γ is not a metric as $\gamma(X, X) \neq 0$.

1.3 Convergence rate

Proposition 2. *Let X and Y be stationary ergodic discrete-time Markov chains, and define*

$$\hat{\gamma}_n := \frac{1}{n} d_{\text{OM}}(X_{1:n}, Y_{1:n}).$$

Then there exists a constant $C > 0$ such that, for all $t \geq 0$ and all n large enough,

$$\mathbb{P}(|\hat{\gamma}_n - \gamma| \geq t) \leq 2 \exp\left(-\frac{t^2 n}{2C}\right).$$

Proof. Let $Z = (X, Y)$, $\Omega = \Sigma^2$, and $\Lambda = \Omega^n$. Then Z is a Markov chain on Ω , and we observe the sample

$$((x_1, y_1), \dots, (x_n, y_n)) \in \Lambda.$$

Define

$$\begin{aligned} f : \Lambda &\longrightarrow \mathbb{R}, \\ ((x_1, y_1), \dots, (x_n, y_n)) &\longmapsto \frac{1}{n} d_{\text{OM}}(x_{1:n}, y_{1:n}). \end{aligned}$$

Let $I = \{i \in \{1, \dots, n\} : x_i \neq y_i\}$, and set

$$M := \max(\sup c_\Sigma, \sup(c_{\text{del}} + c_{\text{ins}})). (\text{a preciser})$$

Then,

$$d_{\text{OM}}(x_{1:n}, y_{1:n}) \leq M|I| = M d_H(x_{1:n}, y_{1:n}),$$

where d_H denotes the Hamming distance, i.e., the number of indices $i \in \{1, \dots, n\}$ such that $x_i \neq y_i$. Let $Z_n = ((X_1, Y_1), \dots, (X_n, Y_n))$ and $Z'_n = ((X'_1, Y'_1), \dots, (X'_n, Y'_n))$. Then

$$\begin{aligned} n |f(Z_n) - f(Z'_n)| &= |d_{\text{OM}}(X_{1:n}, Y_{1:n}) - d_{\text{OM}}(X'_{1:n}, Y'_{1:n})| \\ &\leq d_{\text{OM}}(X_{1:n}, X'_{1:n}) + d_{\text{OM}}(Y_{1:n}, Y'_{1:n}) \\ &\leq M(d_H(X_{1:n}, X'_{1:n}) + d_H(Y_{1:n}, Y'_{1:n})) \\ &\leq 2M d_H(Z_n, Z'_n). \end{aligned}$$

Hence, for any $z, z' \in \Lambda$,

$$|f(z) - f(z')| \leq \sum_{i=1}^n c_i \mathbb{1}_{z_i \neq z'_i}, \quad c_i = \frac{2M}{n}, \quad i = 1, \dots, n.$$

McDiarmid's inequality for Markov chains [Pau15] gives, for all $t \geq 0$,

$$\mathbb{P}(|\hat{\gamma}_n - \mathbb{E}\hat{\gamma}_n| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2 \tau_{\min}}\right) \leq 2 \exp\left(-\frac{2t^2 n}{4M^2 \tau_{\min}}\right),$$

where $\tau_{\min}$ is the dependence/mixing parameter from [Pau15]. Thus the bound holds with $C = 4M^2 \tau_{\min}$ (up to the constant convention in the exponent).

Finally,

$$|\hat{\gamma}_n - \gamma| \leq |\hat{\gamma}_n - \mathbb{E}\hat{\gamma}_n| + |\mathbb{E}\hat{\gamma}_n - \gamma|.$$

Kingman's theorem implies $\hat{\gamma}_n \rightarrow \gamma$ a.s., hence also $\mathbb{E}\hat{\gamma}_n \rightarrow \gamma$. For n large enough,

$$|\mathbb{E}\hat{\gamma}_n - \gamma| \leq \frac{t}{2}.$$

So, for n large enough,

$$\begin{aligned}\mathbb{P}(|\hat{\gamma}_n - \gamma| \geq t) &\leq \mathbb{P}\left(|\hat{\gamma}_n - \mathbb{E}\hat{\gamma}_n| \geq \frac{t}{2}\right) \\ &\leq 2 \exp\left(-\frac{t^2 n}{2C}\right).\end{aligned}$$

□

Assumption 2.

$$\forall t < 0, N^2 \exp\left(-\frac{t^2 n}{2C}\right) \xrightarrow[n \rightarrow \infty]{} 0 \quad (e.g. \log N = o(n)).$$

Lemma 1. For any $\varepsilon > 0$, let

$$\mathcal{E}_{N,n}(\varepsilon) := \left\{ \forall 1 \leq i < j \leq N, |\hat{\gamma}_n(i, j) - \Gamma_{Z_i Z_j}| \leq \varepsilon \right\}.$$

Then

$$\mathbb{P}(\mathcal{E}_{N,n}(\varepsilon)) \xrightarrow[n \rightarrow +\infty]{} 1.$$

Proof. By definition,

$$\mathcal{E}_{N,n}(\varepsilon)^c = \left\{ \exists 1 \leq i < j \leq N : |\hat{\gamma}_n(i, j) - \Gamma_{Z_i Z_j}| > \varepsilon \right\} \subseteq \bigcup_{1 \leq i < j \leq N} \left\{ |\hat{\gamma}_n(i, j) - \Gamma_{Z_i Z_j}| > \varepsilon \right\}.$$

By the union bound,

$$\mathbb{P}(\mathcal{E}_{N,n}(\varepsilon)^c) \leq \sum_{1 \leq i < j \leq N} \mathbb{P}\left(|\hat{\gamma}_n(i, j) - \Gamma_{Z_i Z_j}| > \varepsilon\right) \leq 2 \binom{N}{2} \exp\left(-\frac{\varepsilon^2 n}{2C}\right) \leq N^2 \exp\left(-\frac{\varepsilon^2 n}{2C}\right).$$

Under Assumption 2, the right-hand side tends to 0, hence $\mathbb{P}(\mathcal{E}_{N,n}(\varepsilon)) \rightarrow 1$. □

2 Consistency of some clustering algorithms

We consider a mixture model with K Markov chains. We observe N sequences $X_1, \dots, X_N$, each of length n . Each X_i is generated from a latent label $Z_i \in \{1, \dots, K\}$, where component k corresponds to a stationary ergodic Markov-chain distribution $\mathbb{P}_k$, with weight $0 < w_k < 1$. Let $C_k := \{i, Z_i = k\}$. For $k, k' \in \llbracket 1, K \rrbracket$, define

$$\Gamma_{k,k'} := \gamma(\mathbb{P}_k, \mathbb{P}_{k'}).$$

For $i, j \in \llbracket 1, N \rrbracket$, set

$$\hat{\gamma}_n(i, j) := \frac{1}{n} d_{\text{OM}}(X_i, X_j).$$

Let P^* denote the true partition induced by $(Z_i)_{i=1}^N$, i.e., i and j are in the same block of P^* iff $Z_i = Z_j$.

2.1 Hierarchical clustering

Assumption 3. *Let*

$$\Delta_{\text{in}} := \max_{k \in \llbracket 1, K \rrbracket} \Gamma_{kk}, \quad \Delta_{\text{out}} := \min_{\substack{k, k' \in \llbracket 1, K \rrbracket \\ k \neq k'}} \Gamma_{kk'}.$$

We assume that

$$\eta := \Delta_{\text{out}} - \Delta_{\text{in}} > 0.$$

Definition 3 (Hierarchical agglomerative clustering). *Let L be a linkage, i.e., a distance between subsets of $\{1, \dots, N\}$. Starting from L , we construct a sequence of partitions $(\mathcal{P}_l)_{l=0}^{N-1}$ as follows:*

- **Initialization:**

$$\mathcal{P}_0 := (\{1\}, \dots, \{N\}).$$

- **Inductive step:** for $l \in \llbracket 0, N-2 \rrbracket$, let

$$(A_l, B_l) \in \arg \min_{\substack{C, D \in \mathcal{P}_l \\ C \neq D}} L(C, D).$$

The next partition is obtained by merging A_l and B_l :

$$\mathcal{P}_{l+1} := \left(\mathcal{P}_l \setminus \{A_l, B_l\} \right) \cup \{A_l \cup B_l\}.$$

The procedure terminates at level $N-1$, where

$$\mathcal{P}_{N-1} = \{\{1, \dots, N\}\}.$$

In this section, we focus in particular on single-linkage hierarchical clustering, defined by

$$\forall A, B \subseteq \{1 \dots n\}, \quad L(A, B) := \min_{i, j \in A, B} \hat{\gamma}(i, j).$$

This is equivalent to constructing a graph

$$G_t = (\{1 \dots N\}, E_t)$$

for each $t \geq 0$, where $(i, j) \in E_t$ if and only if $\hat{\gamma}_n(i, j) \leq t$. The associated sequence of partitions is then given by the connected components of G_t , and denoted $\mathcal{P}_{N,n}^{SL}(t)$.

Theorem 1. *Assume that assumptions 1-3 hold. Then, for $t \in]\Delta_{\text{in}}, \Delta_{\text{out}}[$, the partition $\mathcal{P}_{N,n}^{SL}(t)$ satisfies, up to a permutation of labels :*

$$\mathbb{P} \left(\mathcal{P}_{N,n}^{SL}(t) = P^* \right) \xrightarrow{n \rightarrow \infty} 1.$$

Proof. Let $\varepsilon > 0$ such that $t \in]\Delta_{\text{in}} + \varepsilon, \Delta_{\text{out}} - \varepsilon[$. Assume that the event $\mathcal{E}_{N,n}(\varepsilon)$ occurs and $C_k \neq \emptyset$ for all k . Then :

- Let i, j such that $Z_i \neq Z_j$. From assumption 2, $\hat{\gamma}_n(i, j) \geq \Delta_{\text{out}} - \varepsilon > t$. Hence, by definition of G_t , each connected component of G_t is included in a unique cluster.
- Let $k \in \llbracket 1, K \rrbracket$. For all $i, j \in C_k$, $\hat{\gamma}_n(i, j) \leq u < \Delta_{\text{in}} + \varepsilon$. Hence, all pairs of elements of C_k are connected in G_t , and C_k is included in a unique connected component of G_t .

It follows that the connected components of G_t are exactly the clusters C_k . Thus, taking $w_{\min} := \min w_k$:

$$\begin{aligned} \mathbb{P}(\mathcal{P}_{N,n}^{SL}(t) \neq P^*) &\leq \mathbb{P}(\mathcal{E}_{N,n}(\varepsilon)^c) + \mathbb{P}(\exists k \in \llbracket 1, K \rrbracket, C_k = \emptyset) \\ &\leq N^2 \exp\left(-\frac{\varepsilon^2 n}{2C}\right) + (1 - w_{\min})^N \\ &\xrightarrow{n \rightarrow +\infty} 0 \text{ by assumption 2.} \end{aligned}$$

□

Idem pour complete linkage, average linkage...

2.2 K-medoids

Definition 4 (K-medoids (PAM)). *The K-medoids (PAM) estimator is*

$$\hat{M}_{N,n} \in \arg \min_{\substack{M \subseteq \{1, \dots, N\} \\ |M|=K}} \Phi_{N,n}(M), \quad \Phi_{N,n}(M) := \frac{1}{N} \sum_{i=1}^N \min_{m \in M} \hat{\gamma}_n(i, m).$$

The set $\hat{M}_{N,n}$ induces a partition $\hat{P}_{N,n}$ of $\{1, \dots, N\}$ by assigning each index i to its closest medoid. We aim to prove that, up to a permutation of labels,

$$\mathbb{P}(\hat{P}_{N,n} = P^*) \xrightarrow{n, N \rightarrow \infty} 1.$$

Assumption 4. *Let $w_{\max} = \max w_k, w_{\min} = \min w_k$. We assume that :*

$$\Delta_{\text{out}} - \Delta_{\text{in}} > \Delta_{\text{in}} \frac{w_{\max}}{w_{\min}}$$

In what follows, we take

$$0 < \varepsilon < \min\left(\frac{\eta}{2}, \frac{1}{2w_{\min} + w_{\max}}(w_{\min}(\Delta_{\text{out}} - \Delta_{\text{in}}) - w_{\max}\Delta_{\text{in}})\right)$$

We work on the event $\mathcal{E}_{N,n}(\varepsilon)$, and we assume $\forall k \in \llbracket 1, K \rrbracket, C_k \neq \emptyset$.

Lemma 2. *Any minimizer M of the K-medoids criterion $\Phi_{N,n}$ contains exactly one index from each cluster.*

Proof. Let M be a minimizer of the K-medoids criterion, and define

$$\Phi_i(M) := \min_{m \in M} \hat{\gamma}_n(i, m),$$

the contribution of X_i to $\Phi_{N,n}(M)$.

Taking $u = \Delta_{\text{in}} + \varepsilon, v = \Delta_{\text{out}} - \varepsilon$, we have from assumption 1 : $u < v$, and as $\mathcal{E}_{N,n}(\varepsilon)$ holds :

$$\begin{cases} \hat{\gamma}_n(i, j) \leq u & \text{if } Z_i = Z_j, \\ \hat{\gamma}_n(i, j) \geq v & \text{if } Z_i \neq Z_j. \end{cases}$$

Assume that there exists $k \in \llbracket 1, K \rrbracket$ such that the cluster $C_k := \{i \in \llbracket 1, N \rrbracket : Z_i = k\}$ contains no medoid from M . By the pigeonhole principle, there exist $m_1, m_2 \in M$ such that $Z_{m_1} = Z_{m_2} =: k'$ with $k' \neq k$.

Fix any $j \in \llbracket 1, N \rrbracket$ such that $Z_j = k$, and set

$$M' := (M \setminus \{m_2\}) \cup \{j\}.$$

We show that $\Phi_{N,n}(M') < \Phi_{N,n}(M)$. Let $i \in \llbracket 1, N \rrbracket$.

- If $Z_i \notin \{k, k'\}$: let $m \in \arg \min_{m \in M} \hat{\gamma}_n(i, m)$. Since $m \in M'$ as well, we get

$$\Phi_i(M') \leq \hat{\gamma}_n(i, m) = \Phi_i(M),$$

hence $\Phi_i(M) - \Phi_i(M') \geq 0$.

- If $Z_i = k$: since $C_k \cap M = \emptyset$, we have $\Phi_i(M) \geq v$. Moreover, $j \in C_k \cap M'$, so $\Phi_i(M') \leq u$. Therefore,

$$\Phi_i(M) - \Phi_i(M') \geq v - u.$$

- If $Z_i = k'$: if m_2 is not a closest medoid to i in M , then the same closest medoid belongs to M' , and again $\Phi_i(M) - \Phi_i(M') \geq 0$. Otherwise, assume that m_2 is a closest medoid to i in M , and let

$$m^* \in \arg \min_{m \in M'} \hat{\gamma}_n(i, m).$$

Since $m_1 \in M' \cap C_{k'}$, one necessarily has $m^* \in C_{k'}$; otherwise, $m^* \notin C_{k'}$ would imply

$$\hat{\gamma}_n(i, m_1) < \hat{\gamma}_n(i, m^*).$$

Hence, from triangle inequality :

$$\begin{aligned} \Phi_i(M) - \Phi_i(M') &= \hat{\gamma}_n(i, m_2) - \hat{\gamma}_n(i, m^*) \\ &\geq -\hat{\gamma}_n(m_2, m^*) \\ &\geq -u \end{aligned}$$

Summing all contributions, we get : ARNAQUE IL FAUT AJOUTER QUE $|C_k|/N$ tend vers w_k et ε plus petit

$$\begin{aligned} N(\Phi_{N,n}(M) - \Phi_{N,n}(M')) &= \sum_{Z_i \neq k, k'} (\Phi_i(M) - \Phi_i(M')) + \sum_{Z_i = k} (\Phi_i(M) - \Phi_i(M')) + \sum_{Z_i = k'} (\Phi_i(M) - \Phi_i(M')) \\ &\geq 0 + Nw_k(v - u) - Nw_{k'}u \\ &\geq N(w_{\min}(v - u) - w_{\max}u) \\ &= N(w_{\min}(\Delta_{\text{out}} - \Delta_{\text{in}}) - w_{\max}(\Delta_{\text{in}} + \varepsilon)) \\ &> 0 \text{ by definition of } \varepsilon \end{aligned}$$

Hence, $\Phi_{N,n}(M') < \Phi_{N,n}(M)$.

□

Theorem 2 (Consistency of K -medoids). *Assume that Assumptions 1–4 hold. Then, the partition $\hat{P}_{N,n}$ induced by a minimizer $\hat{M}_{N,n}$ of the K -medoids criterion satisfies (up to a permutation of labels) :*

$$\mathbb{P}(\hat{P}_{N,n} = P^*) \xrightarrow{n \rightarrow \infty} 1.$$

Proof. According to lemma 2, if $\mathcal{E}_{N,n}(\varepsilon)$ holds, and $C_k \neq \emptyset$ for all k , then $\hat{P}_{N,n} = P^*$. Hence,

$$\begin{aligned} \mathbb{P} \left(\hat{P}_{N,n} \neq P^* \right) &\leq \mathbb{P} (\mathcal{E}_{N,n}(\varepsilon)^c) + \mathbb{P} (\exists k \in \llbracket 1, K \rrbracket, C_k = \emptyset) \\ &\leq N^2 \exp \left(-\frac{\varepsilon^2 n}{2C} \right) + (1 - w_{\min})^N \\ &\xrightarrow{n \rightarrow +\infty} 0. \end{aligned}$$

□

VERSION PLUS RIGOREUSE (à double checker)

In what follows, we take

$$0 < \varepsilon < \min \left(\frac{\eta}{2}, \frac{1}{2w_{\min} + w_{\max}} (w_{\min}(\Delta_{\text{out}} - \Delta_{\text{in}}) - w_{\max}\Delta_{\text{in}}) \right)$$

We work on the event $\mathcal{E}_{N,n}(\varepsilon)$, and we assume $\forall k \in \llbracket 1, K \rrbracket, C_k \neq \emptyset$.

Lemma 3. *Assume that Assumptions 1–4 hold. Then, on $\mathcal{E}_{N,n}(\varepsilon)$, for all N large enough, any minimizer M of the K -medoids criterion $\Phi_{N,n}$ contains almost surely exactly one index from each cluster.*

Proof. Let M be a minimizer of the K -medoids criterion, and define

$$\Phi_i(M) := \min_{m \in M} \hat{\gamma}_n(i, m),$$

the contribution of X_i to $\Phi_{N,n}(M)$.

Taking $u = \Delta_{\text{in}} + \varepsilon, v = \Delta_{\text{out}} - \varepsilon$, we have from Assumption 1 that $u < v$, and as $\mathcal{E}_{N,n}(\varepsilon)$ holds :

$$\begin{cases} \hat{\gamma}_n(i, j) \leq u & \text{if } Z_i = Z_j, \\ \hat{\gamma}_n(i, j) \geq v & \text{if } Z_i \neq Z_j. \end{cases}$$

Assume that there exists $k \in \llbracket 1, K \rrbracket$ such that the cluster $C_k := \{i \in \llbracket 1, N \rrbracket : Z_i = k\}$ contains no medoid from M . By the pigeonhole principle, there exist $m_1, m_2 \in M$ such that $Z_{m_1} = Z_{m_2} =: k'$ with $k' \neq k$.

Fix any $j \in \llbracket 1, N \rrbracket$ such that $Z_j = k$, and set

$$M' := (M \setminus \{m_2\}) \cup \{j\}.$$

We show that $\Phi_{N,n}(M') < \Phi_{N,n}(M)$. Let $i \in \llbracket 1, N \rrbracket$.

- If $Z_i \notin \{k, k'\}$: let $m \in \arg \min_{m \in M} \hat{\gamma}_n(i, m)$. Since $m \in M'$ as well, we get

$$\Phi_i(M') \leq \hat{\gamma}_n(i, m) = \Phi_i(M),$$

hence $\Phi_i(M) - \Phi_i(M') \geq 0$.

- If $Z_i = k$: since $C_k \cap M = \emptyset$, we have $\Phi_i(M) \geq v$. Moreover, $j \in C_k \cap M'$, so $\Phi_i(M') \leq u$. Therefore,

$$\Phi_i(M) - \Phi_i(M') \geq v - u.$$

- If $Z_i = k'$: if m_2 is not a closest medoid to i in M , then the same closest medoid belongs to M' , and again $\Phi_i(M) - \Phi_i(M') \geq 0$. Otherwise, assume that m_2 is a closest medoid to i in M , and let

$$m^* \in \arg \min_{m \in M'} \hat{\gamma}_n(i, m).$$

Since $m_1 \in M' \cap C_{k'}$, one necessarily has $m^* \in C_{k'}$; otherwise, $m^* \notin C_{k'}$ would imply

$$\hat{\gamma}_n(i, m_1) < \hat{\gamma}_n(i, m^*).$$

Hence, from triangle inequality :

$$\begin{aligned} \Phi_i(M) - \Phi_i(M') &= \hat{\gamma}_n(i, m_2) - \hat{\gamma}_n(i, m^*) \\ &\geq -\hat{\gamma}_n(m_2, m^*) \\ &\geq -u. \end{aligned}$$

Summing all contributions, we get : (with $\hat{w}_k := |C_k|/N$):

$$\begin{aligned} \Phi_{N,n}(M) - \Phi_{N,n}(M') &= \frac{1}{N} \sum_{i=1}^N (\Phi_i(M) - \Phi_i(M')) \\ &= \frac{1}{N} \left(\sum_{Z_i \neq k, k'} (\Phi_i(M) - \Phi_i(M')) + \sum_{Z_i=k} (\Phi_i(M) - \Phi_i(M')) + \sum_{Z_i=k'} (\Phi_i(M) - \Phi_i(M')) \right) \\ &\geq 0 + \hat{w}_k(v - u) - \hat{w}_{k'}u. \end{aligned}$$

Since the labels $(Z_i)_{i=1}^N$ are i.i.d. with $\mathbb{P}(Z_i = k) = w_k$, we have $\hat{w}_k \rightarrow w_k$ almost surely, hence

$$\frac{\min_{\ell} \hat{w}_{\ell}}{\max_{\ell} \hat{w}_{\ell}} \xrightarrow{a.s.} \frac{w_{\min}}{w_{\max}}.$$

By Assumption 4 and by definition of ε , we have :

$$\frac{w_{\min}}{w_{\max}} > \frac{u}{v - u}$$

Therefore, for all N large enough (a.s.),

$$\min_{\ell} \hat{w}_{\ell}(v - u) > \max_{\ell} \hat{w}_{\ell}u,$$

and in particular $\hat{w}_k(v - u) - \hat{w}_{k'}u > 0$. This implies $\Phi_{N,n}(M) > \Phi_{N,n}(M')$, contradicting the optimality of M . Hence, for N large enough (a.s.), every minimizer contains at least one index from each cluster. Since $|M| = K$, it contains exactly one index from each cluster. $\square$

Theorem 3 (Consistency of K -medoids). *Assume that Assumptions 1–4 hold. Then, the partition $\hat{P}_{N,n}$ induced by a minimizer $\hat{M}_{N,n}$ of the K -medoids criterion satisfies (up to a permutation of labels):*

$$\mathbb{P}\left(\hat{P}_{N,n} = P^*\right) \xrightarrow{n \rightarrow \infty, N \rightarrow \infty} 1.$$

Proof. By Lemma 3, on $\mathcal{E}_{N,n}(\varepsilon)$ and for all N large enough, $\hat{P}_{N,n} = P^*$. Hence

$$\begin{aligned} \mathbb{P}(\hat{P}_{N,n} \neq P^*) &\leq \mathbb{P}(\mathcal{E}_{N,n}(\varepsilon)^c) + \mathbb{P}(\exists k \in \llbracket 1, K \rrbracket : C_k = \emptyset) \\ &\leq N^2 \exp\left(-\frac{\varepsilon^2 n}{2C}\right) + (1 - w_{\min})^N \\ &\xrightarrow{n \rightarrow +\infty} 0. \end{aligned}$$

□

Généralisation aux HMM : facile.

2.3 Lower and upper bounds for $\gamma(X, Y)$

Let $\bar{d} : \Sigma_{\perp} \rightarrow \mathbb{R}^+$ defined as :

$$\begin{cases} \forall a, b \in \Sigma, & \bar{d}(a, b) := c_{\text{sub}}(a, b) \\ \forall a \in \Sigma, & \bar{d}(a, \perp) := c_{\text{del}}(a) \\ \forall b \in \Sigma, & \bar{d}(\perp, b) := c_{\text{ins}}(b) \end{cases}$$

Under assumption 1, $\bar{d}$ defines a distance on $\Sigma_{\perp}$.

Lemma 4. *Let X and Y be stationary ergodic discrete-time Markov chains with stationary distributions π_X and π_Y . Let $S := (c_{\text{sub}}(a, b))_{a, b \in \Sigma}$ and let $W_{\bar{d}}$ denote the 1-Wasserstein distance on the finite metric space $(\Sigma_{\perp}, \bar{d})$. Then*

$$W_{\bar{d}}(\pi_X, \pi_Y) \leq \gamma(X, Y) \leq \pi_X^T S \pi_Y$$

Proof. We first prove the lower bound.

Let $f : \Sigma_{\perp} \rightarrow \mathbb{R}$ be a 1-Lipschitz function. Let $x = X_{1:n}$ and $y = Y_{1:n}$, and let

$$(\tilde{x}, \tilde{y}) = ((\tilde{x}_1, \dots, \tilde{x}_L), (\tilde{y}_1, \dots, \tilde{y}_L)) \in \Sigma_{\perp}^L$$

be an alignment of x and y , obtained by inserting $\perp$ symbols so that both sequences have the same length L .

By definition of $\bar{d}$ and since f is 1-Lipschitz, we have

$$\begin{aligned} d_{OM}(x, y) &\geq \sum_{t=1}^L \bar{d}(\tilde{x}_t, \tilde{y}_t) \\ &\geq \sum_{t=1}^L |f(\tilde{x}_t) - f(\tilde{y}_t)| \\ &\geq \left| \sum_{t=1}^L f(\tilde{x}_t) - \sum_{t=1}^L f(\tilde{y}_t) \right| \quad (\text{triangle inequality}). \end{aligned}$$

Since x and y both have length n , each aligned sequence contains exactly n original symbols and $L - n$ inserted $\perp$ symbols. Therefore,

$$\sum_{t=1}^L f(\tilde{x}_t) = \sum_{t=1}^n f(x_t) + (L - n)f(\perp), \quad \sum_{t=1}^L f(\tilde{y}_t) = \sum_{t=1}^n f(y_t) + (L - n)f(\perp),$$

and the $\perp$ terms cancel. Hence

$$d_{OM}(x, y) \geq \left| \sum_{t=1}^n f(x_t) - \sum_{t=1}^n f(y_t) \right|.$$

Taking expectations and dividing by n , we obtain

$$\begin{aligned} \frac{1}{n} \mathbb{E}[d_{OM}(X_{1:n}, Y_{1:n})] &\geq \mathbb{E} \left[\left| \frac{1}{n} \sum_{t=1}^n f(X_t) - \frac{1}{n} \sum_{t=1}^n f(Y_t) \right| \right] \\ &\geq \left| \frac{1}{n} \mathbb{E} \left[\sum_{t=1}^n f(X_t) \right] - \frac{1}{n} \mathbb{E} \left[\sum_{t=1}^n f(Y_t) \right] \right| \quad (\text{Jensen's inequality}). \end{aligned}$$

Since X and Y are stationary,

$$\frac{1}{n} \mathbb{E} \left[\sum_{t=1}^n f(X_t) \right] = \mathbb{E}[f(X_1)], \quad \frac{1}{n} \mathbb{E} \left[\sum_{t=1}^n f(Y_t) \right] = \mathbb{E}[f(Y_1)].$$

Letting $n \rightarrow \infty$, we obtain

$$\gamma(X, Y) \geq |\mathbb{E}_{\pi_X}[f] - \mathbb{E}_{\pi_Y}[f]|.$$

Taking the supremum over all 1-Lipschitz functions f and using the Kantorovich–Rubinstein duality theorem yields

$$\gamma(X, Y) \geq W_{\bar{d}}(\pi_X, \pi_Y).$$

We now prove the upper bound. Let $X_{1:n}, Y_{1:n}$. Taking the sequence of operations consisting of replacing each x_t with y_t , we have :

$$d_{OM}(X_{1:n}, Y_{1:n}) \leq \sum_{t=1}^n \bar{d}(x_t, y_t)$$

By dividing by n , taking expectations, and passing to the limit, we obtain:

$$\gamma(X, Y) \leq \mathbb{E}[\bar{d}(x_1, y_1)]$$

as X and Y are stationary. Hence,

$$\begin{aligned} \gamma(X, Y) &\leq \sum_{a, b \in \Sigma} \pi_X(a) \pi_Y(b) \bar{d}(a, b) \\ &= \pi_X^T S \pi_Y \end{aligned}$$

□

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